

SAM BAI

PRODUCT ENGINEER | 0→1 PRODUCTS & PRODUCTION SYSTEMS

New Zealand | NZ Permanent Resident | +64 27 460 4700 | sambai.codes@gmail.com
sambai.dev | github.com/sambai-dev | linkedin.com/in/sam-bai-dev | Open to Australia / NZ roles

EXPERIENCE

Founder & Product Engineer | Solynt Labs Limited

2026 - Present

Founded a New Zealand software company delivering client systems and owned products.

RIVET | Construction operations platform - client delivery

Jul 2026 - Present

- Shipped a live, private programme and exception platform for a New Zealand residential building team, unifying role-scoped director and project-manager workflows across web, packaged mobile, private calendar feeds and push notifications.
- Engineered conflict-safe local-first sync with IndexedDB outboxes, ETag pulls, scoped authentication, idempotent receipts, advisory locks and transactional Neon/Drizzle writes; served by Hono on Cloudflare Workers.
- Built a working-calendar-aware schedule proposal engine that recalculates at apply time, rejects stale changes and atomically writes dates, audit events, attention items and notifications.
- Protected coordinated releases with migrations, paired API/web smoke tests and automatic web rollback; the current code passes 958 unit tests, with release gates for PostgreSQL, accessibility and two-seat sync.

TREKKY | AI job-search operating system - owned product

Jan 2026 - Present

- Built and operate a live Next.js/React/TypeScript product across web, PWA, an MV3 extension and authenticated MCP, covering discovery, tracking, contacts, follow-ups, AI apply kits, analytics and Google sync.
- Architected atomic scheduled agent jobs with leases, recovery and retries, plus nine-source ingestion across NZ, Australia, the US and Singapore with SHA-256 canonical deduplication and provenance.
- Kept application automation review-first so users submit; cut the dashboard gzip bundle 31.7% (341 KB to 233 KB) through route-level lazy loading and maintained 767 passing tests across 139 files.

SELECTED SYSTEMS

CLEARFOLD | Pre-production digital-asset operations & applied AI

2026

- Built deterministic TypeScript ledger, portfolio and risk packages that enforce balanced entries, idempotent effects and reversals, evidence-linked NAV/P&L and versioned risk policies on React/Vite, Hono and Neon/Drizzle.
- Implemented an evidence-first Gemini assistant with persisted cited conversations, private hybrid retrieval using pgvector plus lexical fallback, and version-bound change proposals; 1,237 tests pass across 150 files.

WORKBENCH | Browser-native product interface

Dec 2025 - Present

- Shipped sambai.dev as an 11-app, three-workspace Workbench with multi-instance windows, focus and z-order, drag, resize, maximise, snap and a responsive compact mode.
- Engineered typed local-first state with versioned atomic commits, revision-conflict detection, schema-validated backup/import and legacy migration, plus an in-browser archive with search, trash and restore.

TECHNICAL SKILLS

Product stack: TypeScript, React, Next.js, Node.js, Vite, Hono, tRPC, Tailwind CSS

Data & systems: PostgreSQL, MongoDB, Redis, Drizzle, local-first sync, workers/queues, RBAC, MCP, hybrid retrieval

Cloud & quality: Cloudflare Workers, R2, Hyperdrive, Vercel, GitHub Actions, Vitest, Playwright, Capacitor, Flutter

EDUCATION

Bachelor of Applied Information Technology (Software Engineering) | Wintec

2024